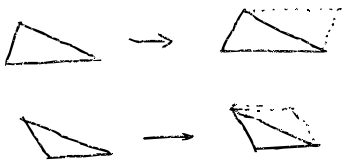
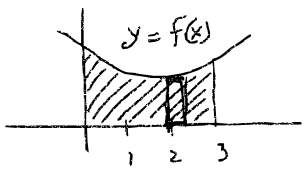
Exercise 4.1



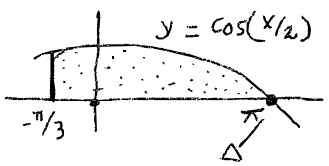
So, $\Delta = \frac{1}{2}\square = \frac{1}{2}bh$

Exercise 4.2.a



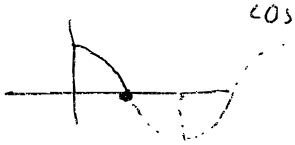
$$\int_0^3 f(x)\,dx$$

Exercise 4.2.b

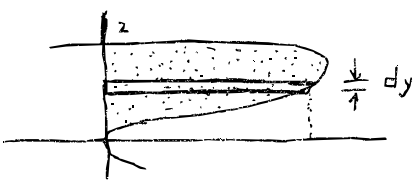


$$\int_{-\pi/3}^{\pi} \cos\left(\frac{x}{2}\right) dx$$

$\cos \frac{x}{2} = 0$



Exercise 4.2.c



$$\int_0^2 (-y^4 + y^3 + 2y^2)\,dy$$

Exercise 4.3.a

No, because $v(t)$ is always positive in the graph

Exercise 4.3.b

Yes, because during 'dt', $v(t)$ is constant. So $v(t)\,dt$ = area of rectangle

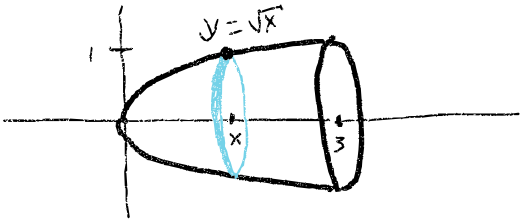


$$\int_4^8 v(t)\,dt$$



$v(t)\,dt$ ← distance traveled in dt

Exercise 4.4.a



$$\int_0^3 \quad dx$$

$$\int_0^3 \pi(\sqrt{x})^2\,dx$$

Because $(x, \sqrt{x})$ will keep distance $\sqrt{x}$ from $(x,0)$ during its 360° rotation around $(x,0)$

That is a circle centered at $(x,0)$ with radius $\sqrt{x}$

Exercise 4.4.b

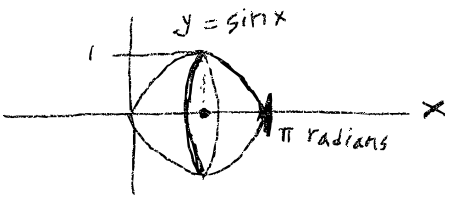
$\pi(\sqrt{x})^2 dx$: volume of a slice at x

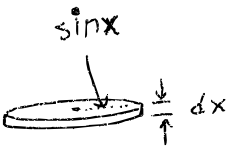
$= \pi x\,dx$

Exercise 4.4.c

$$\int_0^3 \pi(\sqrt{x})^2\,dx$$

Exercise 4.4.d

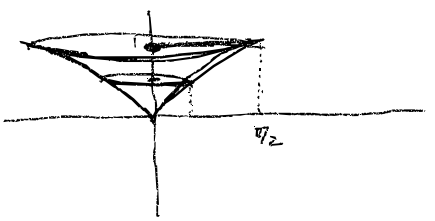




volume of a slice = $\pi(\sin x)^2\,dx$

$$\int_0^{\pi} \pi(\sin x)^2\,dx$$

Exercise 4.4.e

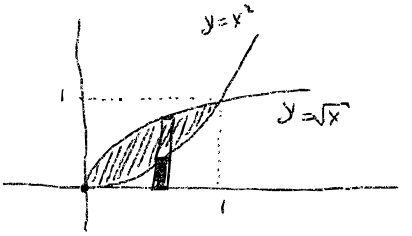


$y = \sin x$

$x = \sin^{-1} y$

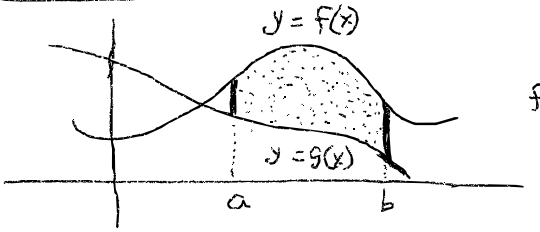
$$\int_0^1 \pi\left(\sin^{-1} y\right)^2 dy$$

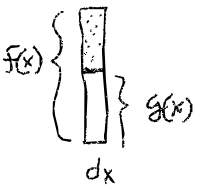
Exercise 4.5.a



$$\int_0^1 (\sqrt{x} - x^2)\,dx$$

Exercise 4.5.b





$$\int_a^b (f(x) - g(x))\,dx$$

Exercise 4.6

$\int_a^b f(x)\,dx = 6$ because in this interval $f(x)$ is never negative. So $f(x)\,dx$ is the area of an infinitely thin rectangular piece. Adding them all up gives the area under the curve between a and b.

Exercise 4.7.a $\int_0^a f(x)\,dx = -5$ f 5 g $5+2+3+1 = 11$

Exercise 4.7.b $\int_a^b f(x)\,dx = 2$

Exercise 4.7.c $\int_0^b f(x)\,dx = -5+2 = -3$

Exercise 4.7.d $\int_a^d f(x)\,dx = 2-3+1 = 0$

Exercise 4.7.e $\int_0^d f(x)\,dx = -5+2-3+1 = -5$

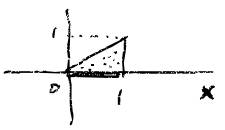
Exercise 4.8.a $\int_0^2 v(t)\,dt$ = distance traveled in first two seconds

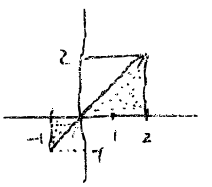
Ex 4.8.b $\int_0^5 v(t)\,dt$ = displacement in first five seconds

Ex 4.8.c $\int_0^5 |v(t)|\,dt$ = distance traveled in first five sec.

Ex 4.8.d $\int_0^2 v(t)\,dt = \int_0^2 |v(t)|\,dt$ (True)

Ex 4.8.e False because left distance > right distance

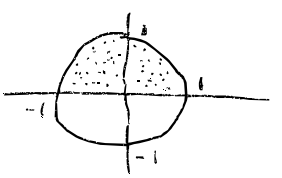
Ex 4.9.a $\int_0^1 x\,dx$ =  $= \frac{1}{2}(1)(1) = \frac{1}{2}$ sq. unit

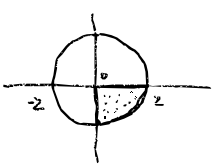
Ex 4.9.b $\int_{-1}^2 2x\,dx$ =  $= 2\left[\frac{1}{2}(2)(2) - \frac{1}{2}(1)(1)\right] = 2\left[2 - \frac{1}{2}\right] = 3$

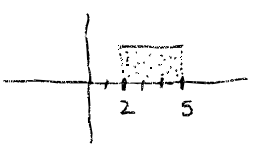
Ex 4.9.c $\int_{-1}^1 \sqrt{1-x^2}\,dx$ $= \frac{1}{2}\pi(1)^2 = \pi/2$

$y^2 = 1 - x^2$

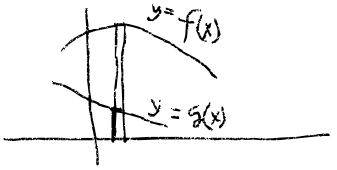
$y = \sqrt{1-x^2}$



Ex 4.9.d $\int_0^2 -\sqrt{4-x^2}\,dx$ =  $= -\frac{1}{4}\pi(2)^2 = -\pi$

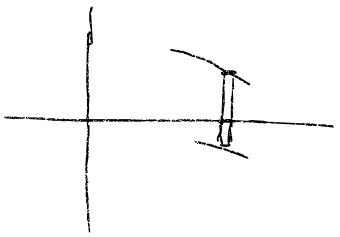
Ex 4.9.e $\int_2^5 1\cdot dx$ =  $= 3$

Ex 4.10.a



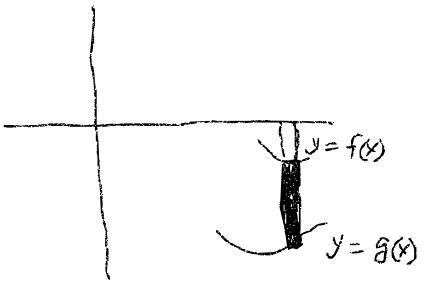
$\left(f(x) - g(x)\right) dx$





$\left(f(x) + [-g(x)]\right) dx$

to make it a length



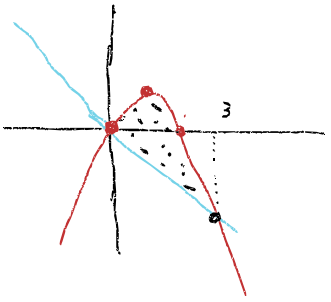
$[-g(x) - (-f(x))]\,dx$

$= [f(x) - g(x)]\,dx$

Ex 4.10.b

Yes $\int_a^c [f(x) - g(x)]\,dx$

Ex 4.10.c



$\int_0^3 [1 - (x-1)^2 + x]\,dx$

$-x = 1 - (x-1)^2$

$x^2 - 2x + 1 - x - 1 = 0$

$= \int_0^3 \left(1 - x^2 + 2x - 1 + x\right) dx$

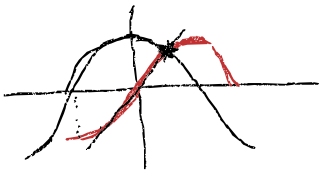
$x^2 - 3x = 0$

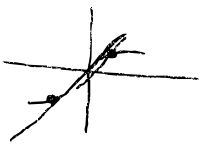
$x(x-3) = 0$

$x = 0, 3$

$= \int_0^3 (3x - x^2)\,dx$

Ex 4.11.a even: x^2, $\cos x$
odd: x, x^3, $\sin x$

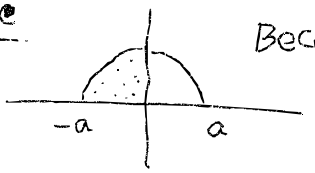




Ex 4.11.b Even symmetric around $x = 0$

Odd Symmetric around $y = x$

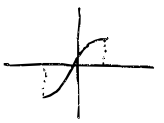
Ex 4.11.c



Because f is symmetric around $x = 0$

So the same area on left and on right of $x = 0$.

Ex 4.11.d Now same area on left and on right of $x = 0$ but sign changed.



Ex 4.11.e $f(x) = -x^3 (\sin(-x))^2 - (-x)\cos(-x)$

Odd

$= -x^3 \sin^2 x + x \cos x$

$= -(x^3 \sin^2 x - x \cos x)$

$= -f(x)$

Ex 4.11.f 0

Ex 4.12.a

$$\frac{d}{dx}\left(cx^{r+1}/r+1\right) = \frac{c}{r+1}(r+1)\,x^{r} = cx^{r}$$ ✓

Ex 4.12.b

$x^3 \to \frac{1}{4}x^4$

$2x^9 \to 2\cdot\frac{1}{10}x^{10}$

$x \to \frac{1}{2}x^2$

$-5\sqrt{x} \to -5\,\frac{1}{3/2}\,x^{3/2} = \frac{10}{3}x^{3/2}$

$\sqrt[3]{x^2} = x^{2/3} \to \frac{1}{5/3}\,x^{5/3}$

$x^{-2} = \frac{1}{-1}x^{-1}$

$-8x^{-3/7} \to -8\,\frac{1}{-\frac{3}{7}+1}\,x^{-\frac{3}{7}+1} = -\frac{56}{4}x^{\frac{4}{7}} = -14x^{4/7}$

$x^{-\pi} \to \frac{1}{1-\pi}x^{1-\pi}$

Ex 4.12.c

$cx^r \neq \frac{d}{dx}\left(cx^{r+1}/r+1\right)$ $(r=-1)$ ✓

$cx^r \neq \frac{c}{r+1}\frac{d}{dx}(x^{r+1})$

$cx^{-1} \to \frac{c}{0}x^0$ ✗

Ex 4.13.a

$\frac{d}{dx}F(x) = \cos x$

$F(x) = \sin x$

Ex 4.13.b

$\frac{d}{dx}F(x) = -\sin x$

$F(x) = \cos x$

Ex 4.13.c

$\frac{d}{dx}F(x) = \sin x$

$F(x) = -\cos(x)$

Ex 4.13.d

$\frac{d}{dx}F(x) = \sec^2 x$

$F(x) = \tan x$

Ex 4.13.e

$\frac{d}{dx}F(x) = e^x$

$F(x) = e^x$

Ex 4.13.f

$\frac{d}{dx}F(x) = \sec x \tan x$

$F(x) = \sec x$

Ex 4.13.g

$$\frac{d}{dx}F(x) = \frac{1}{1+x^2}$$

↑

$\tan^{-1}x$

Ex 4.13.h

$$\frac{d}{dx}F(x) = -\csc^2 x$$

↑

? cotx

Ex 4.13.i

$$\frac{d}{dx}F(x) = \frac{1}{x}$$

↑

$\ln x$

Ex 4.14.a

$$\frac{d}{dx}F(x) = \cos(5x)$$

↓

$$\frac{1}{5}\sin(5x)$$

Ex 4.14.b

$$\frac{d}{dx}F(x) = \sin(\pi x)$$

↓

$$-\frac{1}{\pi}\cos(\pi x)$$

Ex 4.14.c

$$\frac{d}{dx}F(x) = e^{-x}$$

↓

$$-e^{-x}$$

Ex 4.14.d

$$\frac{d}{dx}F(x) = e^{3x}$$

↓

$$\frac{1}{3}e^{3x}$$

Ex 4.14.e

$$\frac{d}{dx}F(x) = \sec^2\left(\frac{x}{2}\right)$$

↓

$$2\tan\left(\frac{x}{2}\right)$$

Ex 4.14.f

$$\frac{d}{dx}F(x) = 10^x$$

↓

$$\frac{1}{\ln(10)}10^x$$

Ex 4.14.g

$$\frac{d}{dx}F(x) = 5^{-x}$$

↓

$$-\frac{1}{\ln(5)}5^{-x}$$

Ex 4.14.h

$$\frac{d}{dx}F(x) = \frac{1}{1+4x^2} = \frac{1}{1+(2x)^2}$$

↓

$$\frac{1}{2}\tan^{-1}(2x)$$

Ex 4.15.a

defined for $x>0$

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

$\ln(-x)$ is defined only for $x<0$

So $\frac{d}{dx}\ln(-x) = -\frac{1}{x} > 0$

They have different domains.

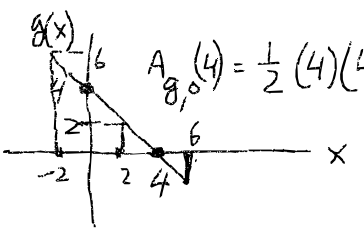
Ex 4.15.b

$g(x) = \frac{1}{x}$ has domain $x \in \mathbb{R}\setminus\{0\}$

$$G(x) = \begin{cases} \ln(x), & x>0 \\ \ln(-x), & x<0 \end{cases}$$

$\ln|x|$

Ex 4.16.a



$A_{g,0}(4) = \frac{1}{2}(4)(4) = 8$

Ex 4.16.b

$A_{g,2}(6) = 0$

Ex 4.16.c

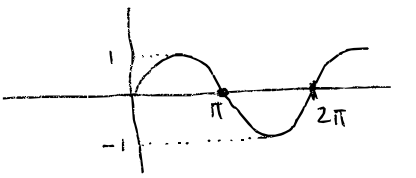
$A_{g,-2}(4) = \frac{1}{2}\cdot 6\cdot 6 = 18$

Ex 4.16.d

$A_{g,1}(1) = 0$

Ex 4.17

$s(x) = \sin x$



$A_{s,0}(2\pi) = 0$

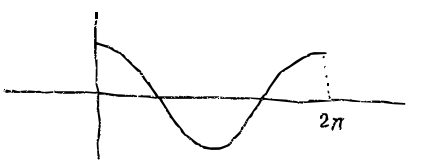
$$A_{f,a}(x) = \int_a^x f(t)\,dt$$

Ex 4.18

$g(x) = \cos(x)$

$$A_{g,0}(x) = \int_0^x g(t)\,dt$$

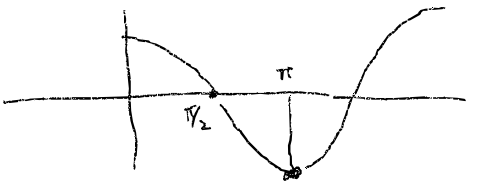
$$= \int_0^x \cos(t)\,dt$$



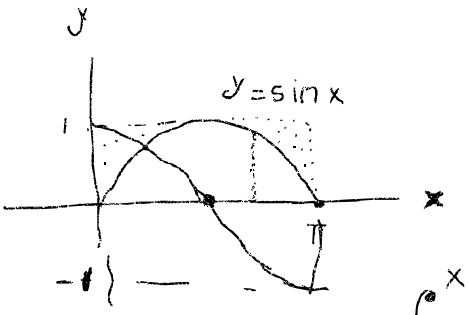
Fundamental theorem of calculus

$f(x)$ —integrate→ $A_{f,a}(x)$ $\left(= \int_a^x f(t)\,dt\right)$

differentiate



Example



$$A_{\sin,0}(x) = \int_0^x \sin(t)\,dt$$

$$A_{\sin,0}(\pi) = \int_0^{\pi} \sin(t)\,dt = ?$$

$$A_{\sin,0}(x) = -\cos(x) + C$$

$$A_{\sin,0}(0) = -\cos(0) + C = 0$$

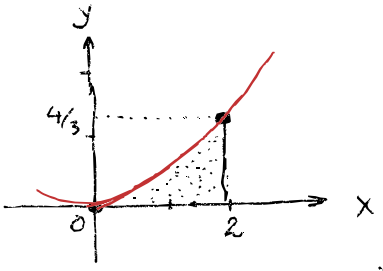
$$C = 1$$

$$A_{\sin,0}(x) = 1 - \cos(x)$$

$$A_{\sin,0}(\pi) = 1 - \cos(\pi) = 1 - (-1) = 2$$

Exercise 19

$y = \frac{1}{3}x^2$



$$A_{y,0}(x) = \int_0^x y(t)\,dt \quad ; \quad A_{y,0}(2) = \int_0^2 \frac{1}{3}x^2\,dx$$

$$= \int_0^x \frac{1}{3}t^2\,dt$$

$$A_{y,0}(x) = \frac{1}{9}x^3 + C$$

$$A_{y,0}(0) = C = 0$$

$$A_{y,0}(x) = \frac{1}{9}x^3$$

$$A_{y,0}(2) = \frac{1}{9}\cdot 2^3 = \frac{8}{9}\ \text{unit}^2$$

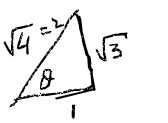
$A_{y,0}(0) = \tan^{-1}(0) + C = 1$

$\tan^{-1}(0) = 1$

so $C = 0$

$A_{y,0}(x) = \tan^{-1} x$

$A_{y,0}(\sqrt{3}) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}\ \text{unit}^2$

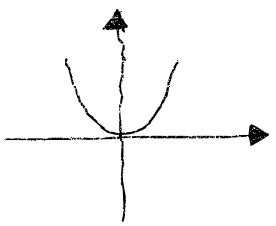


$\tan^{-1}(\sqrt{3}) = \theta$

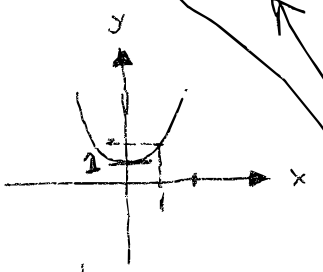
$\sqrt{3} = \tan\theta$

Exercise 20

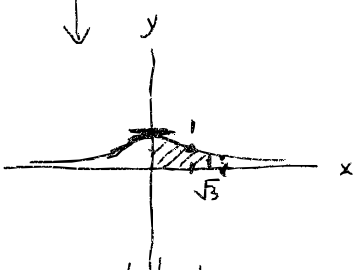
$y = \frac{1}{1+x^2}$



→



$y = 1 + x^2$



$A_{y,0}(x) \equiv$ antiderivative of $y = 1/1+x^2$

One antiderivative of y is $\tan^{-1}x$ because $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

So all antiderivatives of y is like $\tan^{-1}x + C$

$A_{y,0}(x) = \tan^{-1}x + C$

Exercise 21

(a) $\frac{d}{dx}\boxed{\int_a^x f(t)\,dt} = f(x)$

$A_{f,a}(x)$

(b) $\frac{d}{dx}\left(\int_a^x \sin(t)\,dt\right) = \sin(x)$

(c) $\frac{d}{dx}\left(\int_a^x \sin(t^2)\,dt\right) = \sin(x^2)$

(d) $\frac{d}{dx}\left(\int_a^{x^2} \sin(t)\,dt\right) = \boxed{2x\,\sin(x^2)}$ ✓

$y = A_{a,\sin}(x^2)$

$u = x^2$ $\quad \frac{dy}{dx} = \frac{dy}{du}\cdot\left(\frac{du}{dx}\right) = 2x$

$\frac{dy}{du} = \sin(x^2)$

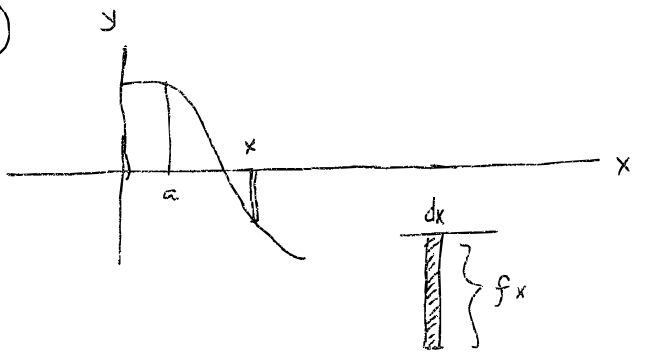
$y = A_{a,\sin}(u)$

$\frac{d}{du}\left(\int_a^u \sin(t)\,dt\right) = \sin(u) = \sin x^2$

Exercise 22

(a)

(b)

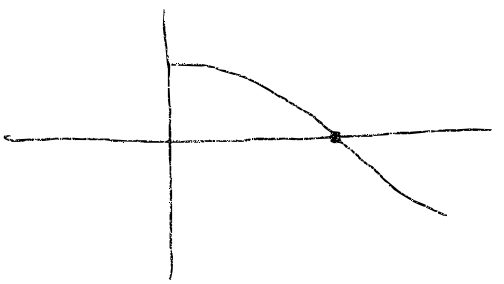


Δ area $= -f(x)\,dx$

$$dA_{f,a}(x) = -(\text{area}) = f(x)\,dx$$

$$\frac{dA_{f,a}(x)}{dx} = f(x)$$

(c)



even then it works because $dA_{f,a}(x) = 0$

which makes left side $= 0$ and right $f(x) = 0$ by def^n.

Chapter 4

(Example 3) $\int_0^2 (-2x^2 + 4x + 1)\,dx$

$$= \left[-2 \cdot \frac{1}{3}x^3 + 4 \cdot \frac{1}{2}x^2 + x\right]_0^2$$

$$= -\frac{2}{3} \cdot 2^3 + 2 \cdot 2^2 + 2$$

$$= -\frac{16}{3} + 8 + 2$$

$$= -\frac{16}{3} + 10$$

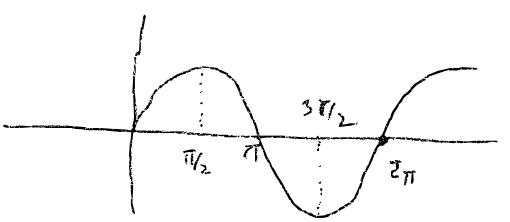
$$= \frac{30 - 16}{3}$$

$$= \frac{14}{3}$$

Exercise 4.23.a

$$\int_0^{3\pi/2} \cos x\,dx$$

$$= \left[\sin x\right]_0^{\frac{3\pi}{2}}$$



$$= \sin\frac{3\pi}{2} - 0$$

$$= -1$$

Exercise 4.23.b

$$\int_1^2 -2x^3\,dx = -2\cdot\left[\frac{1}{4}x^4\right]_1^2$$

$$= -\frac{1}{2}\left(2^4 - 1^4\right)$$

$$= -\frac{1}{2}\cdot 15$$

$$= -15/2$$

Exercise 4.23.c

$$\int_1^{e^2} \frac{1}{x}\,dx = \left[\ln x\right]_1^{e^2} = \ln(e^2) - \ln(1)$$

$$= 2 - 0$$

$$= 2$$

Exercise 4.23.d

$$\int_1^{\sqrt{3}} \frac{1}{1+x^2}\,dx = \left[\tan^{-1} x\right]_1^{\sqrt{3}} = \tan^{-1}(\sqrt{3}) - \tan^{-1}(1)$$

$$= \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{4\pi - 3\pi}{12}$$

$$= \frac{\pi}{12}$$

Exercise 4.23.e

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin x \, dx = -\left[\cos x\right]_{\pi/6}^{\pi/4}$$

$$= -\left(\cos\frac{\pi}{4} - \cos\frac{\pi}{6}\right)$$

$$= \cos\left(\pi/6\right) - \cos\left(\pi/4\right)$$

$$= \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{2}$$

Exercise 4.23.f

$$\int_{-1}^{8} \sqrt[3]{x} \, dx = \frac{1}{1+\frac{1}{3}}\left[x^{\frac{1}{3}+1}\right]_{-1}^{8}$$

$$= \frac{3}{4}\left(8^{4/3} - (-1)^{4/3}\right)$$

$$= \frac{3}{4}\left(16 - 1\right)$$

$$= \frac{45}{4}$$

(Exercise 4.23.g)

$$\int_0^{\frac{\pi}{9}} \cos(3x)\,dx = \left[\frac{1}{3}\sin(3x)\right]_0^{\frac{\pi}{9}}$$

$$= \frac{1}{3}\left(\sin\frac{\pi}{3} - \sin 0\right)$$

$$= \frac{1}{3}\left(\frac{\sqrt{3}}{2} - 0\right)$$

$$= \frac{\sqrt{3}}{6}$$

(Exercise 4.23.h)

$$\int_0^{\sqrt{3}/6} \frac{1}{\sqrt{1-9x^2}}\,dx$$

$$= \frac{1}{3}\left[\sin^{-1}(3x)\right]_0^{\frac{\sqrt{3}}{6}}$$

$$= \frac{1}{3}\left[\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - 0\right]$$

$$= \frac{1}{3}\cdot\frac{\pi}{3}$$

$$= \frac{\pi}{9}$$

Exercise 4.23.i

$$\int_0^1 (x^9 + 5x^4 - 3x)\,dx$$

$$= \left[\frac{1}{10}x^{10} + 5\cdot\frac{1}{5}x^5 - 3\cdot\frac{1}{2}x^2\right]_0^1$$

$$= \frac{1}{10} + 1 - \frac{3}{2}$$

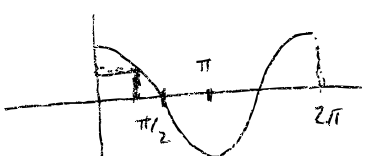
$$= \frac{1 + 10 - 15}{10}$$

$$= -\frac{4}{10}$$

$$= -\frac{2}{5}$$

(Exercise 4.23.j)

$$\int_0^{\frac{\pi}{8}} 2\sin x \cos x \, dx = \int_0^{\frac{\pi}{8}} \sin 2x \, dx$$



$$= -\frac{1}{2}\left[\cos 2x\right]_0^{\frac{\pi}{8}}$$

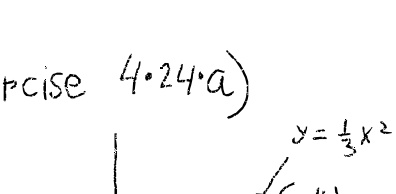
$$= -\frac{1}{2}\left(\cos\frac{\pi}{4} - 1\right)$$

$$= -\frac{1}{2}\left(\frac{\sqrt{2}}{2} - 1\right)$$

$$= -\frac{\sqrt{2}}{4} + \frac{1}{2}$$

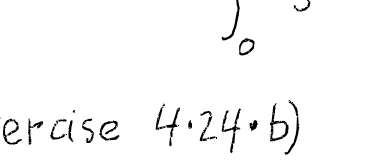
$$= \frac{2-\sqrt{2}}{4}$$

(Exercise 4.24.a)



$$\text{area} = \int_0^2 \frac{1}{3}x^2\,dx = \frac{1}{3}\cdot\frac{1}{3}\left[x^3\right]_0^2 = \frac{1}{9}(8-0) = \frac{8}{9}$$

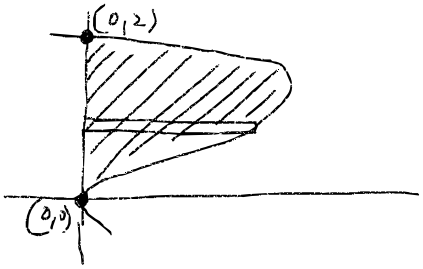
(Exercise 4.24.b)



$$\text{area} = \int_{-\frac{\pi}{3}}^{\pi} \cos\left(\frac{x}{2}\right) dx = 2\left[\sin\left(\frac{x}{2}\right)\right]_{-\pi/3}^{\pi}$$

$$= 2\left[\sin(\pi/2) + \sin(\pi/6)\right] = 2\left(1+\frac{1}{2}\right)$$

(Exercise 4.24.c)



$$-y^4 + y^3 + 2y^2 = 0$$

$$y^2(-y^2 + y + 2) = 0$$

$$y^2 - y - 2 = 0$$

$$y = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} = 2, -1$$

$$\text{area} = \int_0^2 (-y^4 + y^3 + 2y^2)\,dy$$

$$= \left[-\frac{1}{5}y^5 + \frac{1}{4}y^4 + \frac{2}{3}y^3\right]_0^2$$

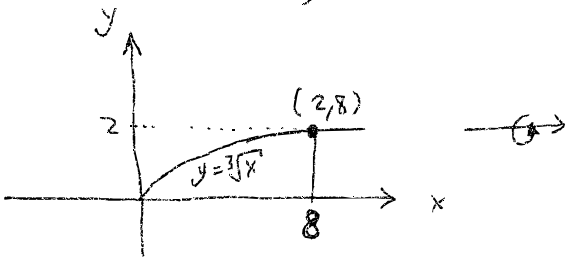
$$= -\frac{1}{5}\cdot 2^5 + \frac{1}{4}\cdot 2^4 + \frac{2}{3}\cdot 2^3$$

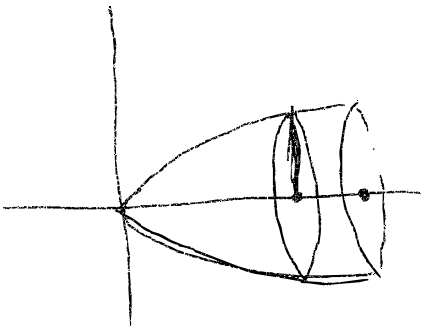
$$= -\frac{32}{5} + 4 + \frac{16}{3}$$

$$= \frac{-96 + 60 + 80}{15}$$

$$= \frac{44}{15}$$

(Exercise 4.25)





total volume = $\int_0^8 \pi x^{\frac{2}{3}}\,dx$

area of slice = $\pi\left(\sqrt[3]{x}\right)^2$

volume of slice = $\pi\ x^{2/3}\,dx$

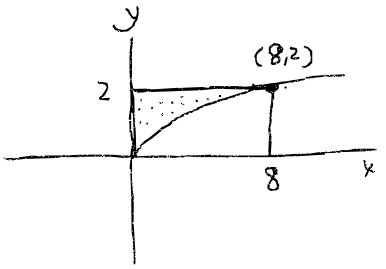
$$= \frac{3\pi}{5}\left[x^{5/3}\right]_0^8$$

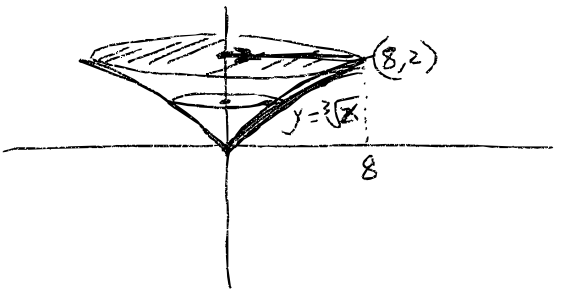
$$= \frac{3\pi}{5} \cdot 8^{5/3}$$

$$= \frac{3\pi}{5} \cdot 32$$

$$= \frac{96\pi}{5}$$

(Exercise 4.26)





$y = \sqrt[3]{x}$

$y^3 = x$

slice area $= \pi(y^3)^2$

$= \pi y^6$

volume of slice $= \pi y^6 \, dy$

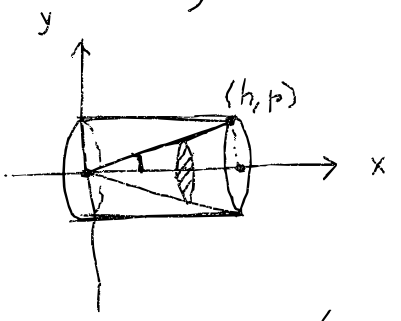
total volume

$$= \int_0^2 \pi y^6 \, dy$$

$$= \frac{\pi}{7} \left[y^7 \right]_0^2$$

$$= \frac{\pi}{7} \cdot 128$$

(Exercise 4.27)



$m = r/h$

$y = \frac{r}{h} x$

slice's area $= \pi \left(\frac{r}{h} x\right)^2$

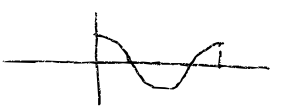
slice's Volume $= \frac{\pi r^2}{h^2} x \, dx$

cone's Volume $= \int_0^h \frac{\pi r^2}{h^2} x^2 \, dx$

$= \frac{\pi r^2}{h^2} \cdot \frac{1}{3} \left[x^3\right]_0^h$

$= \frac{1}{3} \pi r^2 h$

(Exercise 4.28)



$\int_0^{\pi} \sin x \, dx = -\left[\cos x\right]_0^{\pi} = -\cos\pi + 1 = 2$

$\int_{\pi}^{0} \sin x \, dx = \underline{-2}$

(Exercise 4.29)

$$\int_a^b f(x)\,dx - \int_c^b f(x)\,dx + \int_c^d f(x)\,dx$$

$$= \int_a^b + \int_b^c + \int_c^d$$

$$= \int_a^d f(x)\,dx$$

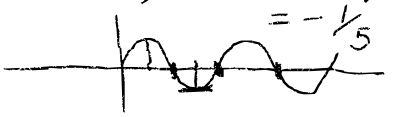
(Exercise 4.30)

(a) -2 (b) 3 (c) 6

(Exercise 4.31)

(a) $\left[c\,F(x)\right]_a^b = c\,F(b) - c\,F(a) = c\left(F(b) - F(a)\right) = c\left[F(x)\right]_a^b$

(b) $\int_0^{3\pi/10} \cos(5x)\,dx = \frac{1}{5}\left[\sin(5x)\right]_0^{3\pi/10} = \frac{1}{5}\left(\sin\frac{3\pi}{2} - 0\right) = \frac{1}{5}\sin\left(\frac{3}{2}\pi\right) = -\frac{1}{5}$



$$\int_0^1 e^{-5x}\,dx = -\frac{1}{5}\left[e^{-5x}\right]_0^1 = -\frac{1}{5}\left(e^{-5} - 1\right) = \frac{1}{5}\left(1 - e^{-5}\right)$$

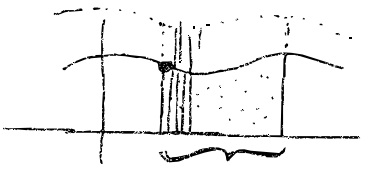
$$\int_4^{25} \sqrt{x}\,dx = \frac{2}{3}\left[x^{3/2}\right]_4^{25} = \frac{2}{3}\left(125 - 8\right) = \frac{234}{3} = 78$$

$$\int_0^{1/4} \frac{1}{1+16x^2}\,dx = \frac{1}{4}\left[\tan^{-1}(4x)\right]_0^{\frac{1}{4}} = \frac{1}{4}\left(\tan^{-1}(1) - \tan^{-1}(0)\right)$$

$$= \frac{1}{4}\left(\frac{\pi}{4} - 0\right) = \pi/16$$

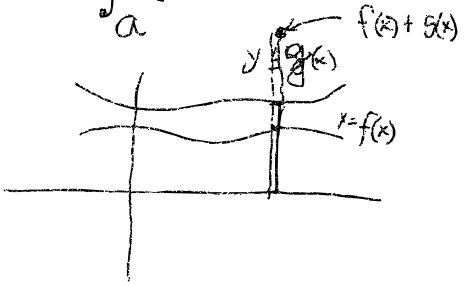
(Exercise 4.32)

(a) $$\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$$



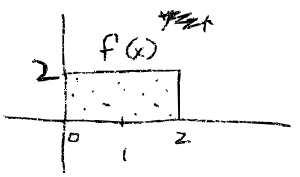
$$\int_a^b c\,f(x)\,dx = A_{c\,f(x),0}(b) - A_{c\,f(x),0}(a)$$

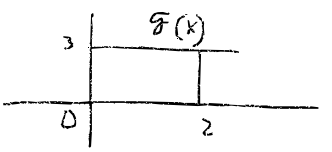
(b) $$\int_a^b [f(x)+g(x)]\,dx$$



(Exercise 4.33)

$$\int_a^b f(x)\,g(x)\,dx \neq \left(\int_a^b f(x)\,dx\right)\left(\int_a^b g(x)\,dx\right)$$





$$\int_0^2 f(x)\,dx = 4 \qquad \int_0^2 g(x)\,dx = 6$$

$$\int_0^2 f(x)\,g(x)\,dx = 12 \neq 24$$

(Exercise 4.34) If f is continuous over $[a,b]$ ...